TONGYE

USER'S MANUAL

☑YDX-100M-35A ☐YDX-100M-35B

SUCTION PUMP

13-2540004

CE
0123

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USERS OF YDX-100 SUCTION PUNP SHOULD BE WELL TRAINED IN THE SAFE AND PROPER USE OF SUCTION EQUIPMENT. THIS MANUAL DOES NOT ATTEMPT TO PROVIDE INFORMATION ON CLINICAL SUCTIONING TECHNIQUES OR INDICATIONS AS TO WHEN TO USE SUCTION. IT'S PURPOSE IS TO GIVE THE USER THE TECHNICAL INFORMATION NECESSARY FOR PROPER OPERATION AND GUIDE HIM IN PERFORMING THE ROUTINE MAINTENANCE PROCEDURES FOR THE YDX-100 SUCTION PUNP.

WARNINGS
BEFORE USING THE APPLIANCE READ THE HANDBOOK

WARNINGS
THE APPLIANCE MUST BE USED BY QUALIFIED PERSONNEL

WARNINGS
DO NOT OPEN THIS DEVICE FOR ANY REASON — PLEASE CONTACT THE MANUFACTURER OR LOCAL DISTRIBUTOR FOR ANY PROBLEM SHOULD ARISE

CAUTIONS
CAREFULLY FOLLOW THE GUIDELINES WHEN USING THE APPLIANCE

DO NOT HANDLE THE APPLIANCE WITH WET OR MOIST HANDS
DO NOT HANDLE THE PLUG WITH WET OR MOIST HANDS
DO NOT USE THE APPLIANCE UNDER THE SHOWER OR IN THE BATHTUB.
DO NOT DIP THE APPLIANCE UNDER WATER
DO NOT USE THE APPLIANCE IN THE PRESENCE OF INFLAMMABLE AND/OR EXPLOSIVE LIQUIDS, GASES OR MIXTURES
THE APPLIENCE SHOULD BE PLACED ON FLAT SURFACE BEING SMALLER THAN 10 DEGREE IN SLOPE WHEN BEING USED.
THE WASTE LIQUID IN THE COLLECTION JAR CAN'T USUALLY EXCEED 70% OF THE VOLUME OF THE COLLECTION JAR AVOIDING TO SUCK THE LIQUID INTO THE SECONDARY ANTI-OVERFLOW BOTTLE AND PUMP. TO TURN OFF MACHINE TIMELY BEFORE THE LIQUID IS FULL AND DISCARD WASTE LIQUID BEFORE FURTHER OPERATION.
DO NOT LEAVE THE APPLIANCE EXPOSED TO ATMOSPHERIC AGENTS OR HEAT SOURCES
DO NOT LET CHILDREN OR INCAPABLE PERSONS USE THE APPLIANCE WITHOUT THE NECESSARY SUPERVISION.
BEFORE STARTING TO CLEAN THE DEVICE, MAKE SURE IT IS NOT CONNECTED TO THE ELECTRIC POWER SUPPLY NETWORK.
DO NOT USE THE APPLIANCE FOR AIMS OTHER THAN THOSE FORESEEN BY THE MANUFACTURER. USING THE APPLIANCE UNDER DIFFERENT CONDITIONS THAN

THOSE SPECIFIED IN THIS HANDBOOK MAY PREJUDICE ITS SAFETY AND ITS TECHNICAL PARAMETERS

INDICATORS

POWER IN	AC 230V Power Input
	Symbol of “Type B”
	Refer to User’s Manual
“ **Class 1** ”	Accessible conductive parts are connected to earth
“ **Type B** ”	Protection against electric shock
“ **Fuse** ”	250V, 2A, Fast Blow Fuse, Dimensions: 5×20 mm.
0123	CE Mark
	Power On
	Power Off
	Protective Earth
	Working switch
	Date of manufacture
	Manufacturer

1. Brief Introduction

Thank you for your employment of the YDX-100 suction pump.

The YDX-100 Suction Pump is used for suction pus, blood, phlegm and extravasate of patients in surgery and first aid(except for thoracic drainage and encephalic drainage).

The YDX-100 Suction pump is high vacuum and high flow equipment.

1. Description of Model

YDX-100□-□□

- (A: vacuometer show vacuum; B: LCD show vacuum)
- Suction capacity: (L/min)
- (M: mobile style; P: portable style)
- Serial number
- Code Name: suction pump

2. Technical Data

MODEL	**YDX-100M-35A**	**YDX-100M-35B**
LENGTH	500MM	500MM
WIDTH	460MM	460MM
HEIGHT	960MM	960MM
WEIGHT	22KG	22KG
POWER SUPPLY	AC230V 50HZ	AC230V 50HZ
POWER CONSUMPTION	150VA	150VA
VACUUM DISPLAY	VACUO METER	LCD
MAX. VACUUM	-0.09MPa	-0.09MPa
MAX. SUCTION CAPACITY	35L/MIN	35L/MIN

FUSE Φ5 ×20/2A Φ5 ×20/2A

4. Installation

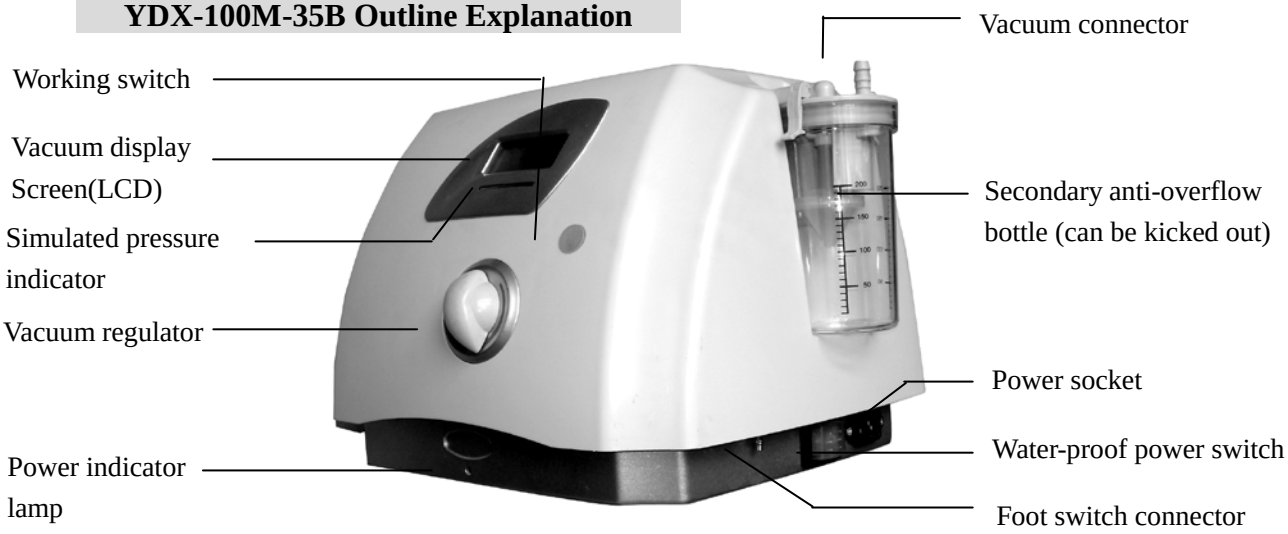
4.1 Packing List

NO.	NAME	SIZE	QUANTITY	REMARK
1	MAIN UNIT		1	
2	POST		1	
3	BASE		1	
4	CASTOR WITHOUT BRAKE		2	
5	CASTOR WITH BRAKE		2	
6	SOCKET HEAD SCREW	M6×15	4	
		M10×30	4	
ACCESSORIES				
7	FOOT SWITCH		1	
8	COLLECTION JAR	3000ML	2	2000ML(OPTIONAL)
9	CONNECTOR OF JAR		6	
10	HOOK FOR JAR		2	
11	FLOATER OF JAR		2	
12	TUBE	Φ8	2 METER	
13	ABDOMINAL SUCTION TUBE		1	
14	FILTER		2	
15	FUSE	Φ5×20/2A	4	Including both ones in Main Unit
16	POWER CABLE		1	
DOCUMENTS				
17	USER’S MANUAL		1	
18	PACKING LIST		1	
TOOLS				

19	SOCKET HEAD SPANNER	5#	1	
		8#	1	

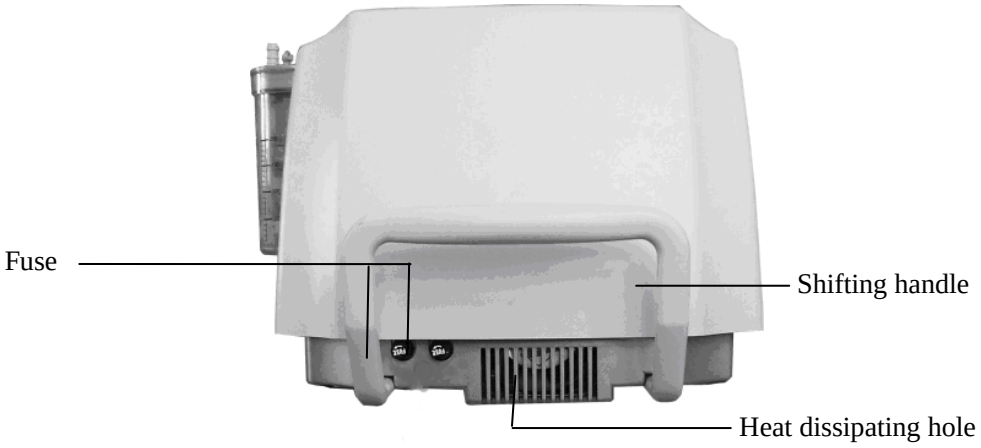
4.2 Outline Explanation

YDX-100M-35B Outline Explanation

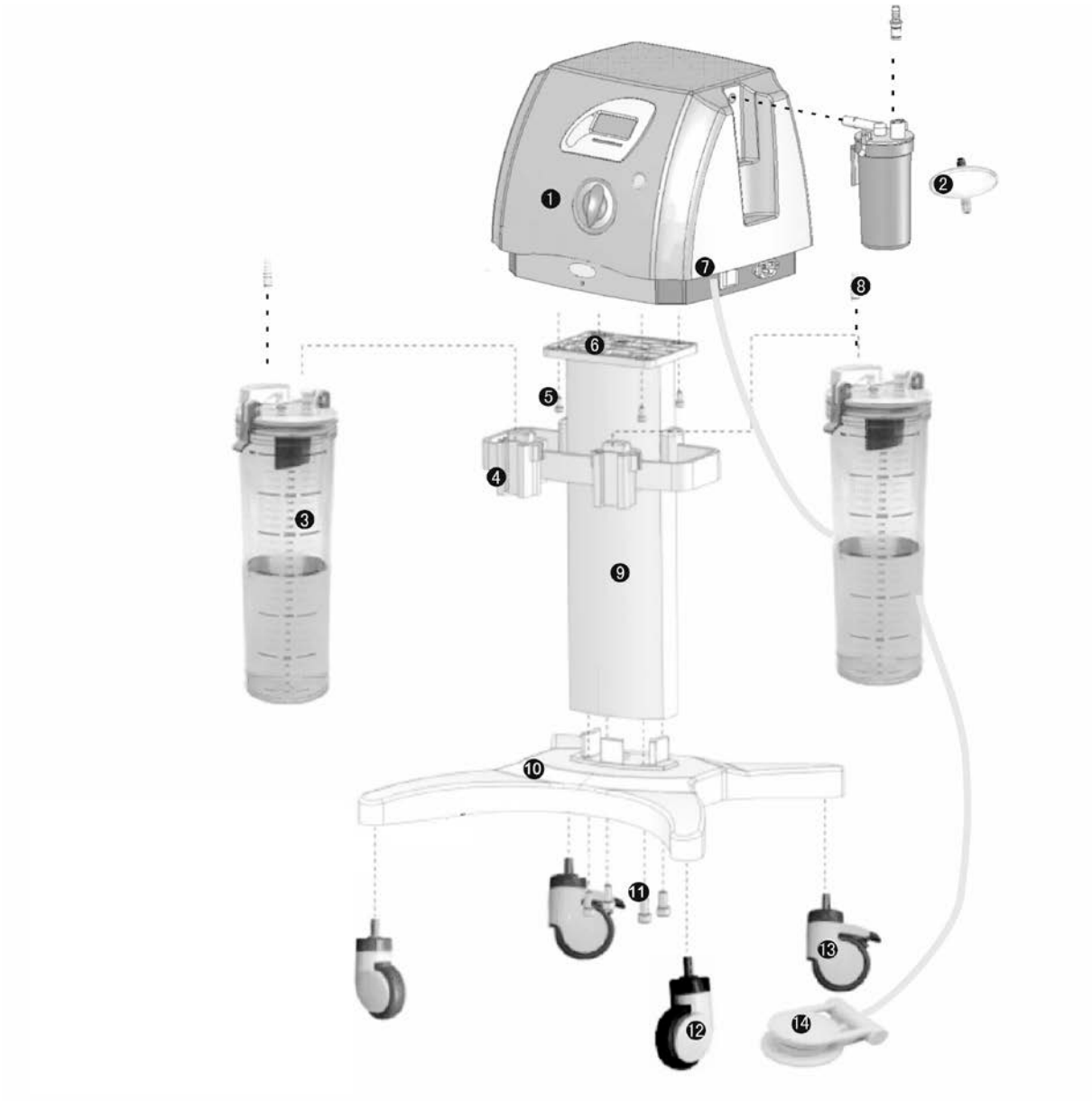


YDX-100M-35A Outline Explanation

BACK

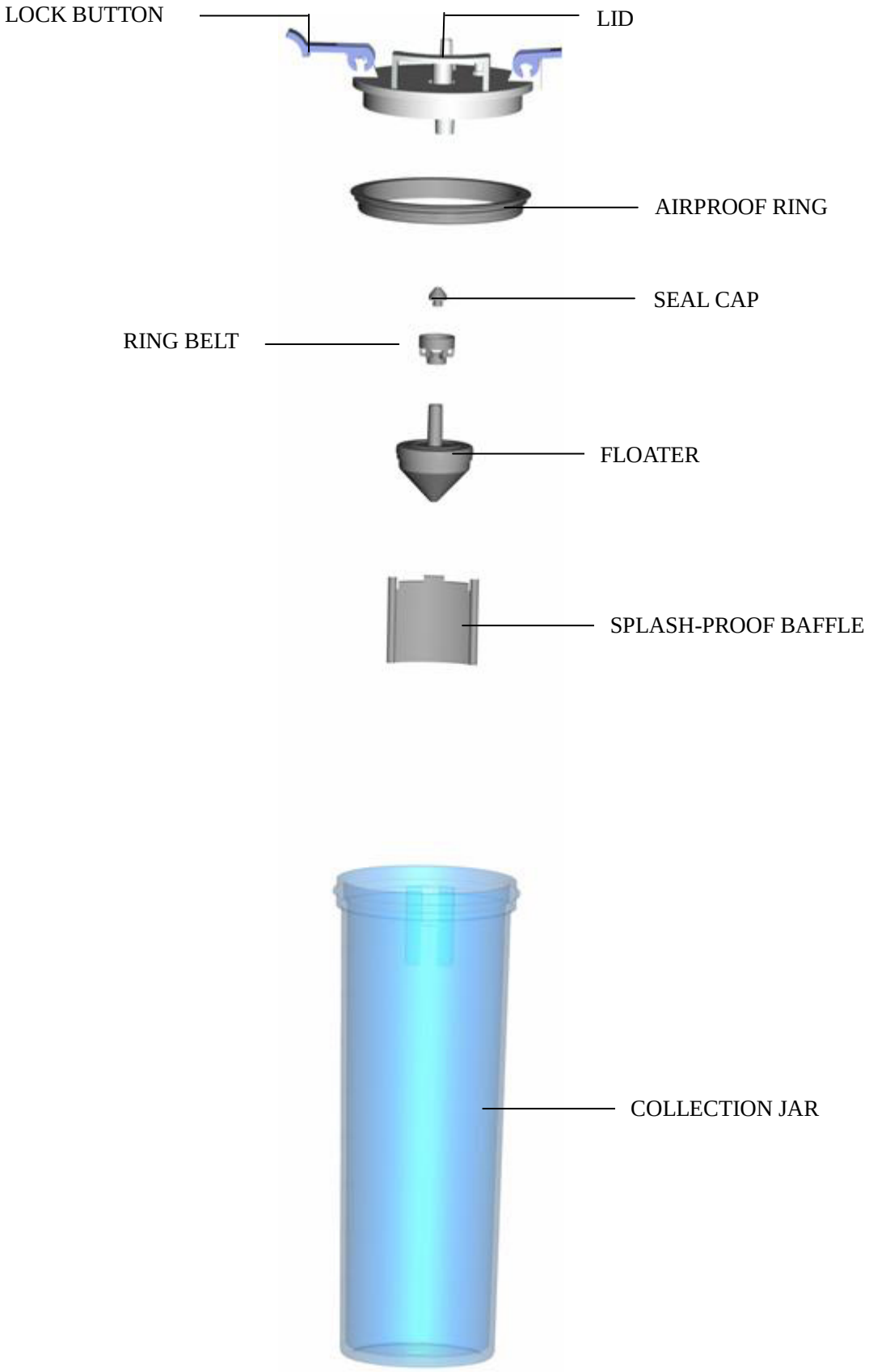


4.3 Installation Diagram



No.	Name	Size	Qty.	No.	Name	Size	Qty.
1	Main unit		1set	8	Connector for jar		3pcs
2	Filter		1pcs	9	Post		1set
3	Collection jar	2000ml(optional) 3000ml	2set	10	Base		1set
4	Hook for jar		2pcs	11	Socket head cap screw	M10×30	4pcs
5	Socket head cap screw	M6×15	4pcs	12	Castors without brake		2pcs
6	Fixing plate		1pcs	13	Castor with brake		2pcs
7	Pipeline joint for foot switch	Φ4	1pcs	14	Foot switch		1pcs

4.4 Installation Diagram of Jar



5. Operation

Operational Temperature: Between 5℃~35℃

Operational Humidity: ≤80%

Beyond 2500Meters of height, the aspiration flow may decrease.

YDX-100M-35A

YDX-100M-35B

1. Tightly press the lid of the anti-overflow bottle and the collection jar, and then turn down the lock button to lock the lid. According to the installation diagram 4.3 to install the connection jars on the appliance.
2. Connect about 100MM tube to the vacuum connector of the anti-overflow bottle; connect the other end of the 100MM tube to the filter.
3. Connect one end of about 600MM tube to the other end of the filter; connect another end of the 600MM tube to the “VACUUM” vent with the floater on the collection jar’s lid.
4. Connect the tube with a suitable length (by operator decide) to the “PATIENT” vent on the connection jar’s lid; connect another end of this tube to the probe.
5. Plug in the power cord to the main but only after checking the voltage is the one indicated by the label in the back of the equipment.
6. Press the working switch.
7. Aspiration intensity can be varied with the vacuum regulator and can be checked on the appliance front panel vacuometer or LCD.
8. It is recommended to carry out a first aspiration cycle with at lease 500ML of water only.
9. The aspirated fluid will flow in the connection jar. Should the amount of fluid exceed the jar’s capacity, the overflow valve will stop aspiration.
10. On the job course, if store liquid bottle suck capability attain to device automatism preventing flooding to stem press stoma. according to Figure C put up work：

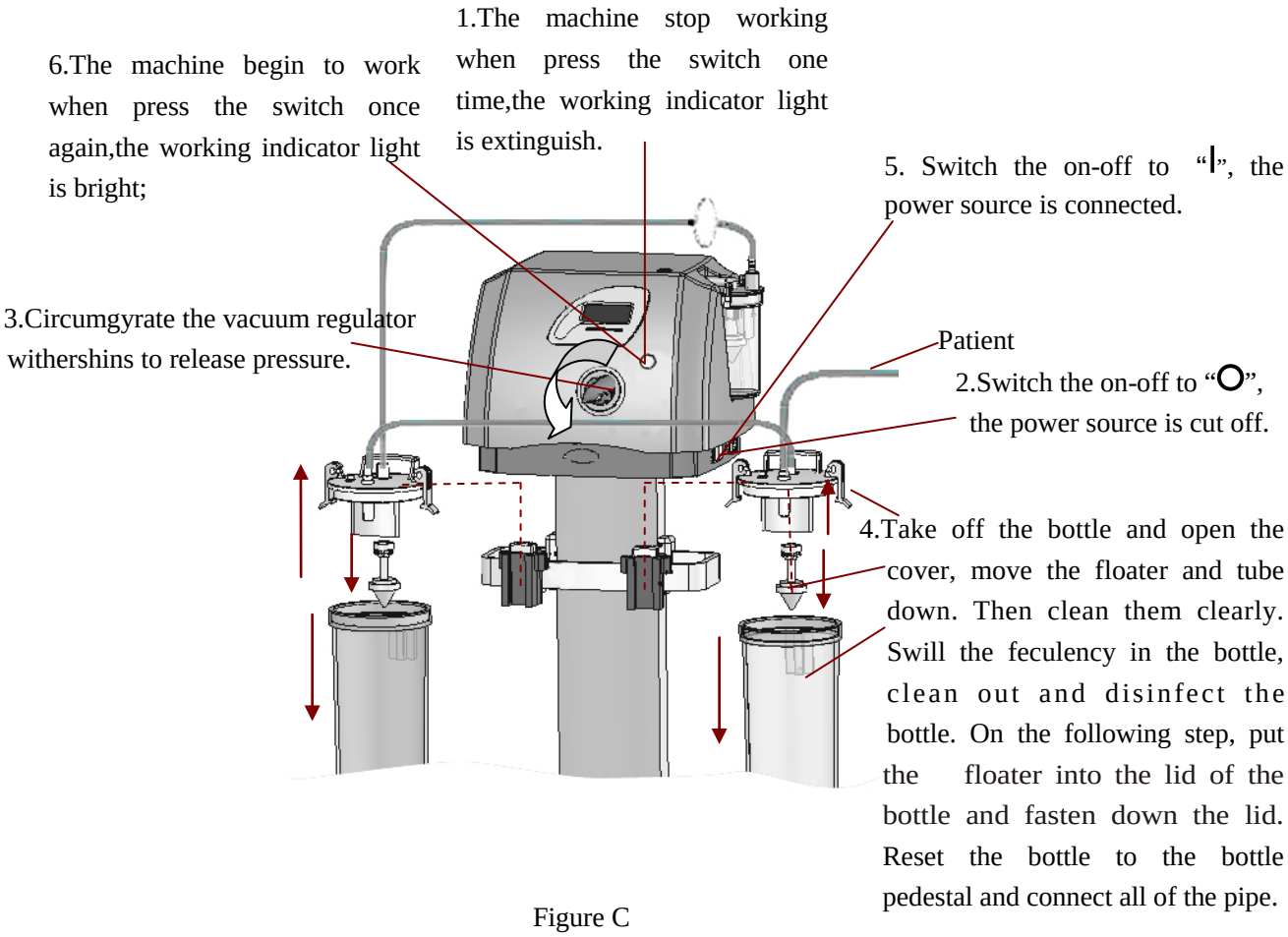
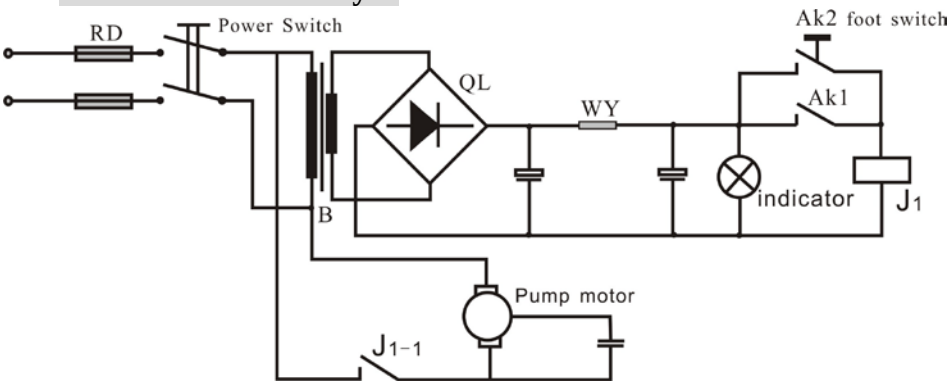


Figure C

CAUTION: Please turn power switch to "Off" when the appliance is not in use.

6. Electrical Circuitry



7. waste water dumpage

A) Turn off the working switch, then lift the collection jar upright and take it far from the suction pump.

B) Open the cover of collection jar, then dump waste water and clean it.

8. Accessories Included

Code No.	Description
YDX022-2000	Collection jar 2000ml

YDX022-3000	Collection jar 3000ml
YDX030	Tube Φ8
YDX011	Antibacterial filter(disposable)

Should accessories be used other than those supplied by Tongye Technologies Development Co.,Ltd, you must make sure that they have the same characteristics as the original ones.

9. Cleaning and Disinfecting

Liquid tube and collection jar should be sterilized and rinsed timely strictly after using.

A) Marinate liquid tube and collection jar in antiseptic solution to sterilize. Max Reuse Limit:30 Cycles.

B) The outside surface of the machine can be cleaned and infected with rag immersed in antiseptic solution.

C) The antiseptic solution can be selected in the light of medical department's liquid tube disinfectant.

D) Three sterilization methods are available, however, it is important to verify that the product can withstand the chosen sterilization method.

1. Moist heat(autoclave)-Please follow EN554 Standards(Sterilization of Medical Device-Validation and Routine Control of Sterilization by Moist Heat).
2. Ethylene Oxide(EO)-Please follow EN550 standards(Sterilization of Medical Devices-Validation and Routine Control of Ethylene Oxide Sterilization).
3. Irradiation(Gamma)-Please follow EN552 Standards(Sterilization of Medical Devices-Validation and Routine Control of Sterilization by Irradiation).

E)

The filter on the appliance is disposable; therefore it is necessary to replace it after every use. Do not use the aspirator without a filter.

10. Disposal

With reference to the disposal of the appliance, its accessories and its single use disposable parts; please see local current waste disposal regulations.

11. periodic Inspection

It is necessary to carry out a daily check by operator before using the appliance.

A) Check for vacuum: make sure the regulator is turned to the high vacuum position.

Then block the vacuum port and observe whether the LCD screen or meter gauge shows the maximum vacuum value. And also check if there's a leak somewhere.

B) Check the dependability of anti-overflow valve.

12. General Troubleshooting

Trouble	Cause	Remedies
Low limit negative value	Bottle not tighten	Tight bottle cap
	Filter seal failure	Replace filter
	Air leakage from pipe joint	Link pipe
	Negative pressure valve failure	Repair or replace by professional persons
	Negative pressure pump leakage	

Indicating lamp fails to light, pump does not work	Socket linkage not reliable	Link socket
	Fuse failure	Replace fuse
	Abnormal volt	Repair or contact the company and distributor
	Pump overload	
	Fluid or solid has been drawn into vacuum pump	
Foot switch does not work	Foot switch wrong linkage	Refer to illustration to link pipe

NOTE: Should any fluid leak in the appliance, call the manufacturer or local distributor for maintenance.

13. Transportation and Store

Packed suction pump shall be placed in room with temperature of 5-40℃ and relative humidity is within 80% and good ventilation and without corruption gas. No upside down and shocking during transportation.

14. Warranty

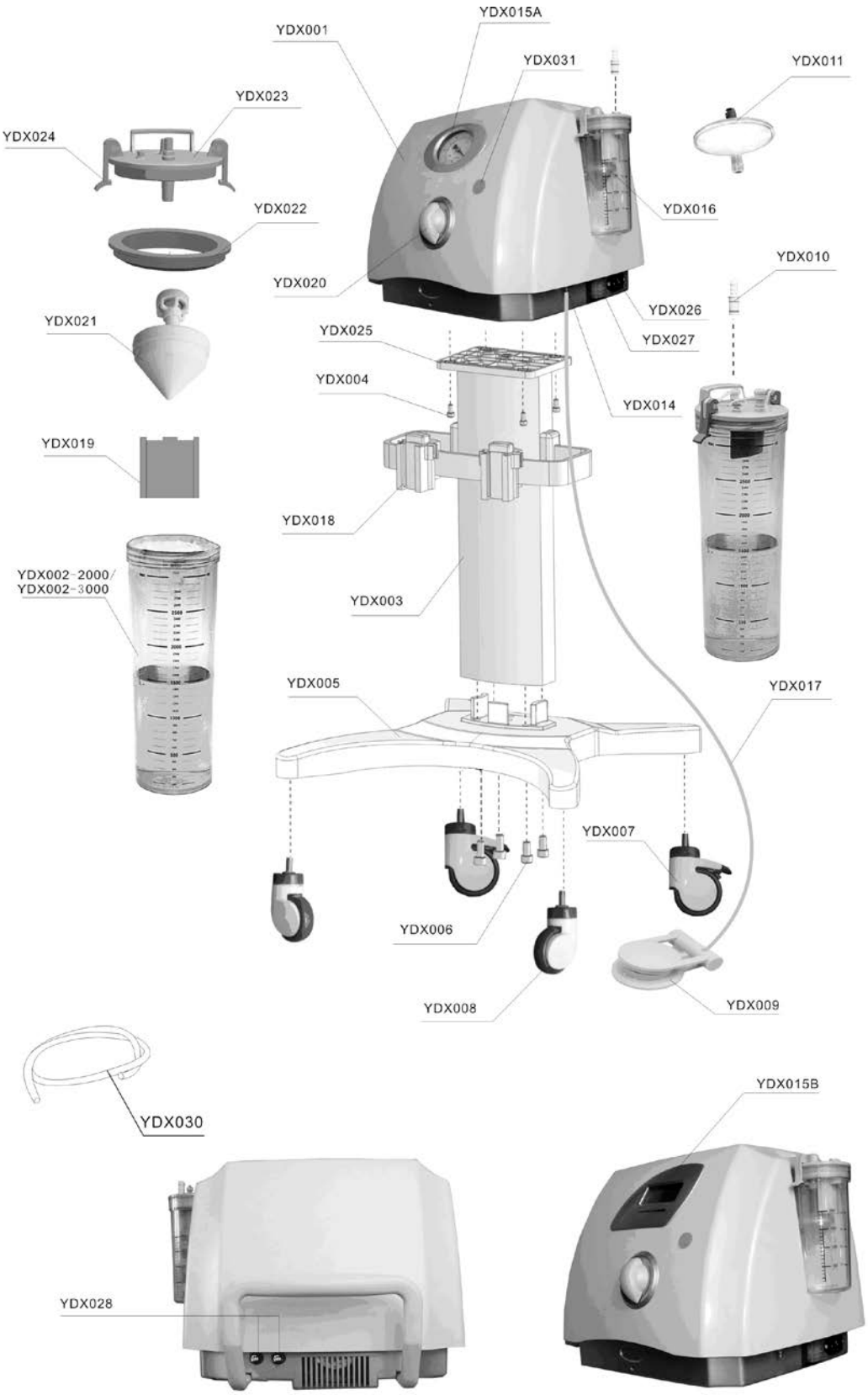
1. The product has a warranty of one year from the date of purchasing, on the premise that it is used in accordance with this Manual.
2. If the instrument does not work properly, please contact the warrantor immediately with the warranty card. The warrantor has the right to refuse repair in case there is no warranty card or the warrantor didn't receive the warranty registration card.
3. Following repairs will be charged within warranty period:
(1) Problems accidents and man-made damages.
(2) Damages caused by unauthorized repair;
(3) Damages caused by inappropriate operation.
4. We provide continuous maintenance and repair after warranty period with certain charges.

TONGYE TECHNOLOGIES DEVELOPMENT CO., LTD RESERVES ALL THE RIGHTS TO CHANGE THIS HANDBOOK WITH NO PRIOR NOTICE FUTURE TO MANUFACTURING CHANGES TO THE APPLIANCE, WHICH HAS BEEN BUILT IN COMPLIANCE WITH THE FOLLOWING STANDARDS: IEC60601-1 AND ISO10079-1
DEVICE IN CONFORMITY WITH 93/42/EEC

Manufacturer: TONGYE TECHNOLOGIES DEVELOPMENT CO., LTD
Address: D2-2-A, Xinmao Science Skill Park. Hua Yuan Industry Development Area, 300384 Tianjin, P.R. China.
URL: http://www.tongye.com.cn Email: sophia@tongye.com.cn
Tel: +86-22-83718080 Fax: 86-22-83716808

EC representative: Shanghai International Holding Corp. GmbH (Europe)
Address: Eiffestrasse 80, 20537 Hamburg, Germany
Tel: 0049-40-2513175 Fax: 0049-40-255726
E-mail: shholding@hotmail.com

YDX-100M-35A/35B SUCTION PUMP COMPONENT


YDX015A
YDX001
YDX031
YDX011
YDX023
YDX024
YDX016
YDX022
YDX010
YDX020
YDX026
YDX021
YDX027
YDX025
YDX004
YDX014
YDX019
YDX018
YDX002-2000/
YDX002-3000
YDX003
YDX005
YDX017
YDX007
YDX006
YDX008
YDX009
YDX030
YDX015B
YDX028

Component No.	Component Name	Description	Component No.	Component Name	Description
YDX001	Main unit		YDX016	Secondary anti-overflow bottle	
YDX002-2000	Collection jar	2000ml	YDX017	Pipeline for foot switch	Φ4
YDX002-3000	Collection jar	3000ml	YDX018	Hook for jar	
YDX003	Post		YDX019	Splash-proof baffle	
YDX004	Socket head cap screw	M6X20	YDX020	Vacuum regulator	
YDX005	Base		YDX021	Floater	
YDX006	Socket head cap screw	M10X30	YDX022	Airproof ring	Φ 119 × Φ108×15
YDX007	Castor with brake	Φ80x20	YDX023	Lid	
YDX008	Castor without brake	Φ80x20	YDX024	Lock button	
YDX009	Foot switch		YDX025	Fixing plate	160 × 120 ×10
YDX010	Connector for jar		YDX026	Power supply socket	
YDX011	Filter		YDX027	Waterproof power switch	
YDX014	Pipeline joint for foot switch		YDX028	Fuse	
YDX015A	vacuometer	Φ60	YDX030	Tube	Φ8
YDX015B	LCD		YDX031	Working switch	

Appendix A:

Guidance and manufacture's declaration – electromagnetic emissions for all EQUIPMENT and SYSTEMS

<table>
<tr><th colspan="3">Guidance and manufacture's declaration – electromagnetic emission</th></tr>
<tr><td colspan="3">The YDX-100M-35A, YDX-100M-35B Suction Pump is intended for use in the electromagnetic environment specified below. The customer of the user of the YDX-100M-35A, YDX-100M-35B Suction Pump should assure that it is used in such an environment.</td></tr>
<tr><th>Emission test</th><th>Compliance</th><th>Electromagnetic environment – guidance</th></tr>
<tr><td>RF emissions
CISPR 11</td><td>Group 1</td><td>The YDX-100M-35A, YDX-100M-35B, Suction Pump uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.</td></tr>
<tr><td>RF emission
CISPR 11</td><td>Class B</td><td rowspan="3">The YDX-100M-35A, YDX-100M-35B, Suction Pump is suitable for use in all establishments, including domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.</td></tr>
<tr><td>Harmonic emissions
IEC 61000-3-2</td><td>Class A</td></tr>
<tr><td>Voltage fluctuations/flicker emissions
IEC 61000-3-3</td><td>Complies</td></tr>
</table>

Guidance and manufacture's declaration – electromagnetic immunity –for all EQUIPMENT and SYSTEMS

Guidance and manufacture's declaration – electromagnetic immunity			
The YDX-100M-35A, YDX-100M-35B Suction Pump is intended for use in the electromagnetic environment specified below. The customer or the user of The YDX-100M-35A, YDX-100M-35B Suction Pump should assure that it is used in such an environment.			
Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment guidance
Electrostatic discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	±6 kV contact ±8 kV air	Floors should be wood, concrete or ceramic tile. If floor are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines	±2kV for power supply lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV differential mode ±2 kV common mode	±1 kV differential mode ±2 kV common mode	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5% UT (>95% dip in UT) for 0.5 cycle 40% UT (60% dip in UT) for 5 cycles 70% UT (30% dip in UT) for 25 cycles <5% UT (>95% dip in UT) for 5 sec	<5% UT (>95% dip in UT) for 0.5 cycle 40% UT (60% dip in UT) for 5 cycles 70% UT (30% dip in UT) for 25 cycles <5% UT (>95% dip in UT) for 5 sec	Mains power quality should be that of a typical commercial or hospital environment. If the user of the YDX-100M-35A, YDX-100M-35B, Suction Pump requires continued operation during power mains interruptions, it is recommended that the YDX-100M-35A, YDX-100M-35B Suction Pump be powered from an uninterruptible power supply or a battery.

Power frequency (50/60Hz) magnetic field IEC61000-4-8	3A/m	3A/m	Power frequency magnetic fields Should be at levels characteristic of a typical location in a typical commercial or hospital environment.
NOTE UT is the a.c. mains voltage prior to application of the test level.			

Recommended separation distances between portable and mobile RF communications equipment and the EQUIPMENT or SYSTEM – for EQUIPMENT or SYSTEM that are not LIFE-SUPPORTING

Recommended separation distances between portable and mobile RF communications equipment and the YDX-100M-35A, YDX-100M-35B Suction Pump.

The YDX-100M-35A, YDX-100M-35B Suction Pump is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the The YDX-100M-35A, YDX-100M-35B Suction Pump can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the YDX-100M-35A, YDX-100M-35B Suction Pump as recommended below, according to the maximum output power of the communications equipment.

Rated maximum output power of transmitter (W)	Separation distance according to frequency of transmitter (m)		
	150 kHz to 80 MHz $d=[3.5 / V_1]\sqrt{p}$	80MHz to 800MHz $d=[3.5 / E_1]\sqrt{p}$	800MHz to2.5GHz $d=[7 / E_1]\sqrt{p}$
0.01	0.1167	0.1167	0.2334
0.1	0.3689	0.3689	0.7378
1	1.1667	1.1667	2.3334
10	3.6893	3.6893	7.3786
100	11.6667	11.6667	23.3334

For transmitters rated at a maximum output power not listed above, the recommended separation distance (d) in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

NOTE1 At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies. NOTE2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

Guidance and manufacture's declaration – electromagnetic immunity –

for EQUIPMENT and SYSTEMS that are not LIFE-SUPPORTING

Guidance and manufacture's declaration – electromagnetic immunity

The YDX-100M-35A, YDX-100M-35B Suction Pump is intended for use in the electromagnetic environment specified below. The customer or the user of the YDX-100M-35A, YDX-100M-35B Suction Pump should assure that it is used in such an environment.

Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment -guidance
Conducted RF IEC 61000-4-6 Radiated RF IEC 61000-4-3	3 Vrms 150 kHz to 80MHz 3 V/m 80 MHz to 2.5GHz	3 Vrms 3 V/m	Portable and mobile RF communications equipment should be used no closer to any part of the YDX-100M-35A, YDX-100M-35B Suction Pump, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance $d=[3.5 / V_1]\sqrt{p}$ $d=[3.5 / E_1]\sqrt{p}$ 80MHz to 800MHz $d=[7 / E_1]\sqrt{p}$ 800MHz to2.5GHz Where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres. Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,[a] should be less than the compliance level in each frequency range.[b] Interference may occur in the vicinity of equipment marked with the following symbol:

NOTE 1：At 80 MHz and 800 MHz, the higher frequency range applies.
NOTE 2：These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

a. Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the YDX-100M-35A, YDX-100M-35B Suction Pump is used exceeds the applicable RF compliance level above, the YDX-100M-35A, YDX-100M-35B Suction Pump should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the YDX-100M-35A, YDX-100M-35B Suction Pump.
b. Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.